

A Models

This table provides a detailed description of the model hyperparameters and components. All the disclosed information can be accessed through Hugging Face model configurations.

Model	Causal	Positional Encoding	Layers	Heads/Layer	d_{model}	h_{dims}	Parameters	Authors
albert-base-v2	✗	absolute (learnable)	12	12	768	3072	11 M	[31]
albert-large-v2	✗	absolute (learnable)	24	16	1024	4096	17 M	[31]
albert-xxlarge-v2	✗	absolute (learnable)	12	64	4096	16384	235 M	[31]
bert-base-uncased	✗	absolute (learnable)	12	12	768	3072	110 M	[14]
bert-large-uncased	✗	absolute (learnable)	24	16	1024	4096	340 M	[14]
distilbert-base-uncased	✗	absolute (learnable)	6	12	768	3072	66 M	[40]
xlm-roberta-base	✗	absolute (learnable)	12	8	768	3072	270 M	[13]
xlm-roberta-large	✗	absolute (learnable)	24	16	1024	4096	550 M	[13]
esm2_t30_150M_UR50D	✗	rotary (RoPE)	30	20	640	2560	150 M	[52]
esm2_t33_150M_UR50D	✗	rotary (RoPE)	33	20	1280	5120	650 M	[52]
protbert	✗	absolute (learnable)	30	16	1024	4096	420 M	[16]
gpt-neo-1.3B	✓	absolute (learnable)	24	16	2048	8192	1.3 B	[7]
gpt-neo-125M	✓	absolute (learnable)	12	12	768	3072	125 M	[7]
opt-2.7b	✓	absolute (learnable)	32	32	2560	10240	2.7 B	[56]
opt-1.3b	✓	absolute (learnable)	24	32	2048	8192	1.3 B	[56]
gpt2	✓	absolute (learnable)	12	12	768	3072	117 M	[9]
gpt2-large	✓	absolute (learnable)	36	20	1280	5120	774 M	[9]
gpt2-medium	✓	absolute (learnable)	24	16	1024	4096	345 M	[9]
gpt2-xl	✓	absolute (learnable)	48	25	1600	6400	1.5 B	[9]

Table 1: Architectural details for all model families analyzed. d_{model} = hidden/embedding size; h_{dims} = intermediate (FFN) size.

B Independence of Empirical Results

The following table includes the results from regressing the results from the three experiments conducted.

	Coef	Std err	t	p	Significance
Intercept	2.05	0.10	0	1	ns
CKA	-0.89	0.10	-8.73	$p < 0.01$	***
GW	-0.17	0.10	-1.66	0.1	ns

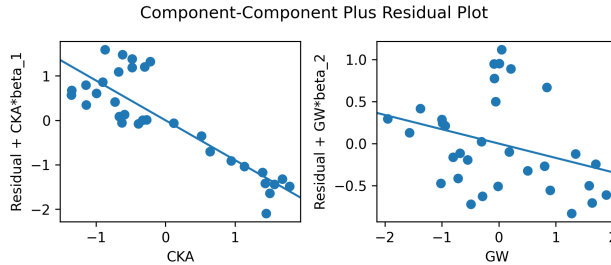


Figure 6: Results for the OLS regression of the experimental results. Adjusted $R^2 = 0.71$.

C Data and Code Availability

The code and data used during the experiments are made available here. The GitHub repository includes instruction on how to proceed in order to reproduce the disclosed results.

Code and datasets available upon acceptance.